

# Middle Convolution of $\ell$ -adic Perverse Sheaves

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# 1 Introduction

## 1.1 Notations

**1.1.1.** Motivated by the growing interest in the formalization of mathematics within the mathematical community, we adopt a style of notation inspired by type theory throughout this write-up, without committing to type-theoretic foundations. The symbols below denote types and/or categories, depending on your philosophical inclination. The notation  $X : T$  means that  $X$  is a term of type  $T$  or an object of the category  $T$ .

|                                      |                                                                                                                                                                                                     |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\text{Set}$                         | Sets                                                                                                                                                                                                |
| $\text{Set}^{\text{fin}}$            | Finite sets                                                                                                                                                                                         |
| $\text{Top}$                         | Topological spaces                                                                                                                                                                                  |
| $\text{Open}(T)$                     | Open subsets of $T : \text{Top}$                                                                                                                                                                    |
| $\text{Grp}$                         | Groups                                                                                                                                                                                              |
| $\text{Grp}^{\text{top}}$            | Topological groups                                                                                                                                                                                  |
| $\text{Grp}^{\text{pro}}$            | Profinite groups                                                                                                                                                                                    |
| $\text{Grp}^{\text{ab}}$             | Abelian groups                                                                                                                                                                                      |
| $\text{Ring}$                        | Rings (associative, unital, not necessarily commutative)                                                                                                                                            |
| $\text{Ring}^{\text{top}}$           | Topological rings                                                                                                                                                                                   |
| $\text{Field}$                       | Fields                                                                                                                                                                                              |
| $\text{Field}^{\text{top}}$          | Topological fields                                                                                                                                                                                  |
| $\text{Mod}_R$                       | Left modules over $R : \text{Ring}$                                                                                                                                                                 |
| $\text{Mod}_R^{\text{top}}$          | Topological left modules over $R : \text{Ring}^{\text{top}}$                                                                                                                                        |
| $\text{Vect}_k$                      | Vector spaces over $k : \text{Field}$                                                                                                                                                               |
| $\text{Vect}_k^{\text{fin}}$         | Finite-dimensional vector spaces over $k : \text{Field}$                                                                                                                                            |
| $\text{Rep}_\Lambda(G)$              | Continuous $\Lambda$ -linear representations of $G : \text{Grp}^{\text{top}}$ , where $\Lambda : \text{Ring}^{\text{top}}$                                                                          |
| $\text{Rep}_\Lambda^{\text{fin}}(G)$ | Continuous $\Lambda$ -linear representations of $G : \text{Grp}^{\text{top}}$ , where $\Lambda : \text{Ring}^{\text{top}}$ , whose underlying topological $\Lambda$ -modules are finitely generated |
| $\text{Sch}$                         | Schemes                                                                                                                                                                                             |
| $\text{Sch}_S$                       | Schemes over a base scheme $S : \text{Sch}$                                                                                                                                                         |
| $\text{Var}_k$                       | Varieties over $k : \text{Field}$ , i.e. integral separated schemes of finite type over $k$                                                                                                         |
| $\text{Curv}_k$                      | Curves over $k : \text{Field}$ , i.e. one-dimensional varieties over $k$                                                                                                                            |
| $\text{Curv}_k^{\text{sm}}$          | Smooth curves over $k : \text{Field}$                                                                                                                                                               |
| $\text{Riem}$                        | Riemann surfaces, i.e. connected one-dimensional complex manifolds                                                                                                                                  |
| $\text{Site}$                        | Sites, i.e. categories equipped with Grothendieck topologies                                                                                                                                        |
| $\text{Sh}(X, A)$                    | Sheaves on $X : \text{Site}$ with values in a category $A$ admitting small limits                                                                                                                   |
| $\text{Mod}(X, \mathcal{R})$         | Sheaves of left $\mathcal{R}$ -modules on $X : \text{Site}$ , where $\mathcal{R} : \text{Sh}(X, \text{Ring})$                                                                                       |
| $\text{D}^b(A)$                      | Bounded derived objects of an abelian category $A$                                                                                                                                                  |
| $\text{D}^b(X, k)$                   | Bounded derived objects of $\text{Sh}(X, \text{Vect}_k)$ , where $X : \text{Site}$ and $k : \text{Field}$                                                                                           |

$X \rightarrow Y$  Morphisms from  $X : T$  to  $Y : T$

## 1.2 Acknowledgements

**1.2.1.** The author thanks his advisor, Prof. Chris Hall, for his guidance and advice.

**1.2.2 Typesetting.** The typesetting style of this write-up is inspired by [Kat88].

**1.2.3 Declaration of AI Use.** The use of generative AI in the preparation of this write-up was limited to the following purposes: polishing wording, phrasing, grammar, and formatting; offering writing advices; answering mathematical questions from the author to help him learn the subject; searching for and citing appropriate references; translating reading materials in French into English; retypesetting old references with archaic typesetting for ease of reading; providing emotional support for the author; providing fitness advice to help the author stay healthy and energetic while working on the write-up.

## 1.3 Motivations

**1.3.1.** *Monodromy groups* are interesting mathematical objects that arise naturally in both geometric and arithmetic contexts. Roughly speaking, they describe how locally defined data change when transported along a loop. Controlling and understanding monodromy groups has been a central theme in both of these worlds, with significant applications.

**1.3.2.** The goal of this write-up is to provide an exposition on monodromy groups as well as ways to study them using methods from perverse sheaves and their middle convolution introduced by Katz [Kat96], especially in the  $\ell$ -adic arithmetic context.

**1.3.3.** Consider the equation

$$z(w)^2 = w \tag{1.3.3.1}$$

for the unknown holomorphic function  $z : V \rightarrow \mathbb{C}^\times$ , where  $V : \text{Open}(\mathbb{C}^\times)$  and  $w : V$  is the independent variable. Fix a base point  $w_0 : \mathbb{C}^\times$  and choose  $z_0 : \mathbb{C}^\times$  with  $z_0^2 = w_0$ . On a sufficiently small disc  $V$  containing  $w_0$ , the equation has two holomorphic solution branches  $z_+(w) = \sqrt{w}$  and  $z_-(w) = -\sqrt{w}$  with  $z_+(w_0) = z_0$  and  $z_-(w_0) = -z_0$ . Move  $w$  once around 0 along the loop

$$\gamma : [0, 1] \rightarrow \mathbb{C}^\times, \quad t \mapsto w_0 e^{2\pi i t}.$$

Along this loop, the values of the two solution branches vary continuously as  $\tilde{\gamma}_+(t) = z_0 e^{\pi i t}$  and  $\tilde{\gamma}_-(t) = -z_0 e^{\pi i t}$ , until they eventually swap places at  $t = 1$ . In other words, there is a permutation representation of the fundamental group<sup>1</sup>  $\pi_1^{\text{top}}(\mathbb{C}^\times, w_0) \cong \mathbb{Z} : \text{Grp}^{\text{top}}$

$$\rho : \pi_1^{\text{top}}(\mathbb{C}^\times, w_0) \rightarrow \text{Aut}_{\text{Set}}(\{[z_+]_{w_0}, [z_-]_{w_0}\}), \quad n \mapsto ([z_+]_{w_0} [z_-]_{w_0})^n.$$

We call this representation the *permutation monodromy representation*, and we call its image, isomorphic to  $S_2$ , the *permutation monodromy group*, of the equation (1.3.3.1).

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<sup>1</sup>with the discrete topology

**1.3.4.** We reformulate the above with sheaves. Consider the covering

$$f : \mathbb{C}^\times \rightarrow \mathbb{C}^\times, \quad \zeta \mapsto \zeta^2.$$

Let  $\mathcal{F} : \text{Sh}(\mathbb{C}^\times, \text{Set})$  be its sheaf of holomorphic sections. Equivalently, this is the sheaf of solutions to the equation (1.3.3.1),

$$\begin{aligned} \mathcal{F}(V) &= \{z : V \rightarrow \mathbb{C}^\times \mid z \text{ holomorphic and } (f \circ z)|^V = \text{id}_V\} \\ &= \{z : V \rightarrow \mathbb{C}^\times \mid z \text{ holomorphic and } z(w)^2 = w \text{ for every } w : V\}. \end{aligned}$$

As in 1.3.3, on a sufficiently small disc  $V$  containing  $w_0$ , the sections are precisely the two holomorphic solution branches  $z_+$  and  $z_-$ , identifying

$$\mathcal{F}|_V \cong \underline{\{+, -\}}_V.$$

Hence  $\mathcal{F}$  is locally constant, with stalk at  $w_0$

$$\mathcal{F}_{w_0} = \{[z_+]_{w_0}, [z_-]_{w_0}\} \cong f^{-1}(w_0)$$

identified by evaluation. However, the sheaf is not constant:  $\mathcal{F}(\mathbb{C}^\times) = \emptyset$ . A global section would determine a germ at  $w_0$  fixed by all monodromy, whereas transport once around 0 exchanges the two square-root germs, as demonstrated in 1.3.3. Locally constant (but not necessarily constant) sheaves of sets, such as this one, are known as *local systems of sets*.

**1.3.5.** For a path  $\gamma : [0, 1] \rightarrow \mathbb{C}^\times$  from  $w_0 = \gamma(0)$  to  $w_1 = \gamma(1)$  and a point  $\zeta : f^{-1}(w_0)$  in the fibre, let  $\tilde{\gamma}_\zeta$  be the unique lift through  $f$  satisfying  $\tilde{\gamma}_\zeta(0) = \zeta$ . We have the *transport map*

$$T_\gamma : \mathcal{F}_{w_0} \xrightarrow{\sim} \mathcal{F}_{w_1}, \quad \zeta \mapsto \tilde{\gamma}_\zeta(1).$$

with the evaluation identifications  $\mathcal{F}_w \cong f^{-1}(w)$ . Thus we obtain a *transport functor*<sup>2</sup>

$$T : \Pi_1(\mathbb{C}^\times) \longrightarrow \text{Set}, \quad w \longmapsto \mathcal{F}_w, \quad [\gamma] \longmapsto T_\gamma.$$

Restricting transport to loops at  $w_0$  gives the same representation in 1.3.3

$$\rho : \pi_1^{\text{top}}(\mathbb{C}^\times, w_0) \rightarrow \text{Aut}_{\text{Set}}(\mathcal{F}_{w_0}), \quad [\gamma] \mapsto T_\gamma.$$

**1.3.6.** In § 2, we will see how to upgrade permutation monodromy representations into linear monodromy representations, which we will just call monodromy representations. Moreover, one can use this machinery to talk about monodromy in the  $\ell$ -adic arithmetic context, by replacing  $\pi_1^{\text{top}}$  with the étale fundamental group  $\pi_1^{\text{ét}}$ . When  $X = \text{Spec}(k)$  for some  $k : \text{Field}$ , with a choice of separable closure  $k^{\text{sep}}$ , then  $\pi_1(X, \bar{x}) = \text{Gal}(k^{\text{sep}}/k)$ , where  $x = \text{Spec}(k^{\text{sep}})$ , is the absolute Galois group of  $k$ . Hence, a monodromy representation in this case degenerates into a Galois representation, a ubiquitous arithmetic object.

<sup>2</sup>Here  $\Pi_1(X)$  denotes the fundamental groupoid of some  $X : \text{Top}$ , that is, a category whose objects are points of  $X$  and whose morphisms are homotopy classes of paths relative to their endpoints, with concatenation as composition.

## 2 Local Systems and Lisse Sheaves

### 2.1 Local Systems in Topology

**2.1.1 Notation.** We use the following notation:

|                                                                                  |                                                       |
|----------------------------------------------------------------------------------|-------------------------------------------------------|
| $X : \text{Curv}_{\mathbb{C}}^{\text{sm}}$                                       | A smooth projective connected curve over $\mathbb{C}$ |
| $S : \text{Set}^{\text{fin}}$                                                    | A finite nonempty subset of closed points of $X$      |
| $U : \text{Curv}_{\mathbb{C}}^{\text{sm}}$                                       | The open subvariety $X \setminus S$                   |
| $j : U \rightarrow X$                                                            | The inclusion map                                     |
| $m : \mathbb{N}$                                                                 | The cardinality of $S$                                |
| $(-)^{\text{an}} : \text{Curv}_{\mathbb{C}}^{\text{sm}} \rightarrow \text{Riem}$ | The complex analytification functor                   |

**2.1.2.** We have looked at permutation monodromy, but permutation actions do not capture all the situations we wish to study, since a lot of times in nature we encounter actions on vector spaces instead of sets. For example, instead of an algebraic equation like the one in 1.3.3, we consider a differential equation, which is known as *Gauss's hypergeometric equation*

$$\lambda(1-\lambda) \frac{d^2 f}{d\lambda^2} + (c - (a+b+1)\lambda) \frac{df}{d\lambda} - abf = 0 \quad (2.1.2.1)$$

where  $a, b, c : \mathbb{C}$  are fixed parameters and  $f : V \rightarrow \mathbb{C}$  is an unknown holomorphic function, with  $V : \text{Open}(U)$ ,  $U = \mathbb{P}^1(\mathbb{C}) \setminus \{0, 1, \infty\}$ , and independent variable  $\lambda : V$ . We then define the sheaf of solutions to this equation  $\mathcal{L} : \text{Sh}(U, \text{Vect}_{\mathbb{C}})$

$$\mathcal{L}(V) = \{f : V \rightarrow \mathbb{C} \mid f \text{ holomorphic and satisfies (2.1.2.1) on } V\}.$$

This is a sheaf of vector spaces since (2.1.2.1) is a homogeneous linear differential equation. For each  $\lambda_0 : U$ , the existence-and-uniqueness theorem [HTT08, Theorem 4.3.1] gives a sufficiently small open disc  $V : \text{Open}(U)$  containing  $\lambda_0$  and an isomorphism of sheaves

$$\underline{\mathbb{C}}_V^2 \xrightarrow{\sim} \mathcal{L}|_V$$

Thus  $\mathcal{L}$  is locally constant of rank 2. We call such a sheaf a *complex local system of rank 2*.

**2.1.3.** Similarly, analytic continuation (A.1.1) gives a *transport functor*

$$T : \Pi_1(U) \rightarrow \text{Vect}_{\mathbb{C}}^{\text{fin}}, \quad \lambda \mapsto \mathcal{L}_{\lambda}, \quad [\gamma] \mapsto T_{\gamma},$$

where, for a path  $\gamma : [0, 1] \rightarrow U$ ,

$$T_{\gamma} : \mathcal{L}_{\gamma(0)} \rightarrow \mathcal{L}_{\gamma(1)}, \quad [f]_{\gamma(0)} \mapsto [f^{\gamma}]_{\gamma(1)}.$$

Here  $[f^{\gamma}]_{\gamma(1)}$  is the solution germ obtained by analytically continuing  $[f]_{\gamma(0)}$  along  $\gamma$ . Restricting transport to loops at a base point  $\lambda_0 : U$  gives the *monodromy representation*

$$\rho : \pi_1^{\text{top}}(U, \lambda_0) \rightarrow \text{Aut}_{\text{Vect}_{\mathbb{C}}}(\mathcal{L}_{\lambda_0}), \quad [\gamma] \mapsto T_{\gamma}.$$

and its image is called the *monodromy group*.

**2.1.4.** Like in 1.3.4,  $\mathcal{L}$  is also a sheaf of sections of a cover.

## 2.2 Local Systems in Complex Algebraic Geometry

## 2.3 Local Systems in $\ell$ -adic Arithmetic Geometry: Lisse Sheaves

## 2.4 Rigidity of Local Systems

# 3 Perverse Sheaves and Middle Convolution

## 3.1 Six-Functor Formalism

## 3.2 Middle Extension Sheaves

## 3.3 Middle Convolution

# 4 Mod- $\ell$ Monodromy of Hyperelliptic Curves

# Appendix

## A.1 Analytic Continuation

### A.1.1. Definition of Analytic Continuation.

# References

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